A Mathematical Study of Convergence of Selected Real Sequences.

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January 12, 2026

Declaration

I hereby declare that this project entitled ***"A Mathematical Study of Convergence of Selected Real Sequences"*** is a self-initiated work carried out independently by me and has not been submitted, in whole or in part, for any other degree or diploma.

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Objective

The objective of this project is to investigate the convergence behaviour of selected real sequences using analytical limit techniques, rigorous ε–N proofs, numerical approximations, and graphical observations.

Introduction

Convergence is a fundamental concept in mathematics that describes how sequences and series approach a specific value as they progress. It is an essential idea in real analysis and forms the basis of limits, continuity, differentiation, and integration.

Convergence ensures that infinite processes, such as series and iterative methods, lead to finite and mathematically meaningful results. This allows mathematicians and scientists to make accurate approximations and to analyze the stability and reliability of mathematical models used in calculus, physics, engineering, and computer science.

For example, in **Fourier series**, periodic functions are expressed as **infinite sums of sine and cosine functions**. Such representations play a key role in areas such as *signal processing* and the study of *heat equations.*

In this project, selected real sequences are studied to illustrate convergence using analytical, numerical, and graphical methods.

Theory & Definitions

Sequence:

Definition: A **sequence of real numbers** (or a **sequence in** $\mathbb{R}$) is a function defined on the set $\mathbb{N} = \{1, 2, \ldots\}$ of natural numbers whose range is contained in the set $\mathbb{R}$ of real numbers.

In other words, a sequence in $\mathbb{R}$ assigns to each natural number $n = 1, 2, \ldots$ a uniquely determined real number.

If $S : \mathbb{N} \to \mathbb{R}$ is a sequence, we denote the value of S at n by the symbol $\boldsymbol{s_n}$ rather than using the functional notation $S(n)$.

Since the domain for a sequence is always $\mathbb{N}$, a sequence is specified by the values s_n, $n \in \mathbb{N}$ and is denoted as

$$(s_n)_{n\in\mathbb{N}} \quad or \quad (s_1, s_2, s_3, \ldots)$$

The values s_n are also called the **terms** or the

elements of the sequence.

The limit of a sequence: ($\varepsilon - N$ **definition**)

Definition: A sequence $S = (s_n)$ in $\mathbb{R}$ is said to **converge** to $L \in \mathbb{R}$, or L is said to be a **limit** of (s_n), if for every $\varepsilon > 0$, there exists a natural number $N(\varepsilon)$ such that for all $n \geq N(\varepsilon)$, the terms s_n satisfy $|s_n - L| < \varepsilon$.

If a sequence has a limit, we say that the sequence is **convergent**; if it has no limit, we say the sequence is **divergent**.

Note:

1. The notation $N(\varepsilon)$ is used to emphasize that the choice of N depends on the value of ε. However, it is often convenient to write N instead of $N(\varepsilon)$.

2. When a sequence has limit L, we will use the notation,

$$\lim_{n\to\infty} S = L \quad or \quad \lim_{n\to\infty} (s_n) = L$$

$\star$ We will sometimes use the symbolism $s_n \to L$ to indicate the intuitive idea that the terms s_n approach the number L as $n \to \infty$.

$\#$ Remark:

The definition of the limit of a sequence of real numbers is used to verify that a proposed value L is indeed the limit. It does not provide a means for initially determining what that value of L is.

Bounded sequence:

Definition: A sequence $S = (s_n)$ of real numbers is said to be bounded if there exists a real number $M > 0$, such that $|s_n| \leq M$, for all $n \in \mathbb{N}$.

Thus, the sequence (s_n) is bounded if and only if the set $\{s_n : n \in \mathbb{N}\}$ of its values is a bounded subset of $\mathbb{R}$.

Theorems:

Theorem 1:
A convergent sequence of real numbers is bounded.

Proof : Suppose that $\lim(s_n) = l$ and $\varepsilon = 1$.

Then there exists a natural number $N = N(1)$ such that $|s_n - l| < 1$ for all $n \geq N$.

If we apply the Triangle Inequality[1] with $n \geq N$, we obtain,

$$|s_n| = |s_n - l + l| \leq |s_n - l| + |l| < 1 + |l|$$

If we set,

$$M = \sup\{|s_1|, |s_2|, \ldots, |s_{N-1}|, 1 + |l|\}$$

then it follows that,

$$|s_n| \ \leq \ M, \text{ for all } n \in N$$

$$Q.E.D$$

[1] *If* $a, b \in \mathbb{R}$, *then* $|a+b| \leq |a| + |b|$.

Theorem 2:
A sequence cannot converge to more than one limit.

Proof : Suppose, if possible, a sequence (s_n) converges to two real numbers l and l' $(l \neq l')$.

Let us select $\varepsilon = \frac{1}{3}\,|l - l'| > 0$.

Since the sequence (s_n) converges to l and l'; therefore, there exist positive integers m_1 and m_2 such that

$$|s_n - l| < \varepsilon, \ \forall \ n \geq m_1 \tag{1}$$

and

$$|s_n - l'| < \varepsilon, \ \forall \ n \geq m_2 \tag{2}$$

Now from (1) and (2), for $n \geq \max(m_1,\ m_2)$

$$|l - l'| = |l - s_n + s_n - l'| \leq |l - s_n| + |s_n - l'| < 2\varepsilon$$

i.e, $|l - l'| < \frac{2}{3}\,|l - l'|$, which is not possible.

Hence, the sequence cannot converge to two limits.

$$Q.E.D$$

Theorem 3: (Sandwich / Squeeze Theorem)
If (a_n), (b_n) and (c_n) are three sequences such that

(i) $a_n \leq b_n \leq c_n, \;\; \forall n$ (1)

(ii) $\lim a_n = \lim c_n = l$

then

$$\lim b_n = l$$

Proof : Let $\varepsilon > 0$ be given.

Now since (a_n),(c_n) both converges to l, therefore $\exists$ positive integers m_1, m_2, such that

$$|a_n - l| < \varepsilon, \quad \forall\, n \geq m_1 \tag{2}$$

and

$$|c_n - l| < \varepsilon, \quad \forall\, n \geq m_2 \tag{3}$$

Let $m = \max(m_1,\; m_2)$.

then, for $n \geq m$, we have from (1),(2) and (3)

$$l - \varepsilon < a_n \leq b_n \leq c_n < l + \varepsilon$$

$$\Rightarrow \qquad l - \varepsilon < b_n < l + \varepsilon,\; \forall\, n \geq m$$

$$\Rightarrow \qquad |b_n - l| < \varepsilon,\; \forall\, n \geq m.$$

Hence,

$$\lim b_n = l$$

$$Q.E.D$$

Theorem 4: (Algebra of Sequences)
If $(a_n), (b_n)$ be two sequences such that $\lim a_n = a, \lim b_n = b$, then

(i) $\lim (a_n \pm b_n) = \lim a_n \pm \lim b_n = a \pm b$

(ii) $\lim (a_n b_n) = (\lim a_n)(\lim b_n) = ab$

(iii) $\lim(a_n/b_n) = (\lim a_n)/(\lim b_n) = a/b$, if $b \neq 0, b_n \neq 0, \ \forall \ n$.

Proof :
(i) Let $\varepsilon > 0$ be given.

Since $\lim a_n = a$ and $\lim b_n = b$, therefore, $\exists$ positive integers m_1 and m_2, respectively, such that

$$|a_n - a| < \frac{1}{2}\varepsilon, \ \forall \ n \geq m_1, \text{ and}$$

$$|b_n - b| < \frac{1}{2}\varepsilon, \ \forall \ n \geq m_2.$$

Thus, for $m = \max(m_1, m_2)$, we have

$$|a_n - a| < \frac{1}{2}\varepsilon, \ |b_n - b| < \frac{1}{2}\varepsilon, \ \forall \ n \geq m$$

$$\begin{aligned} \therefore \qquad |(a_n \pm b_n) - (a \pm b)| &= |(a_n - a) \pm (b_n - b)| \\ &\leq |a_n - a| + |b_n - b| \\ &< \frac{1}{2}\varepsilon + \frac{1}{2}\varepsilon = \varepsilon, \ \forall \ n \geq m \end{aligned}$$

Hence, $\qquad \lim(a_n \pm b_n) = a \pm b = \lim a_n \pm \lim b_n$

(ii) The two sequences $(a_n), (b_n)$ being convergent, are bounded so that $\exists$ positive real numbers k, K such that

$$|a_n| \leq k, |b_n| \leq K, \ \forall \ n$$

Now

$$\begin{aligned} |a_n b_n - ab| &= |a_n(b_n - b) + b(a_n - a)| \\ &\leq |a_n| \cdot |b_n - b| + |b| \cdot |a_n - a| \\ &\leq k\,|b_n - b| + |b| \cdot |a_n - a| \qquad (1) \end{aligned}$$

Let $\varepsilon > 0$ be given,
Since $\lim a_n = a, \lim b_n = b$, therefore there exists positive integers m_1 and m_2, respectively, such that

$$|a_n - a| < \frac{\frac{1}{2}\varepsilon}{|b| + 1}, \ \forall \ n \geq m_1$$

$$|b_n - b| < \frac{\frac{1}{2}\varepsilon}{k}, \ \forall \ n \geq m_2$$

Then, for $m = \max(m_1, m_2)$, we have from (1)

$$|a_n b_n - ab| < \frac{1}{2}\varepsilon + \frac{|b| \cdot \frac{1}{2}\varepsilon}{|b| + 1} < \varepsilon, \ \forall \ n \geq m.$$

Hence, $\qquad \lim(a_n b_n) = ab = (\lim a_n)(\lim b_n).$

(iii) *Lemma.* To show that if $\lim b_n = b \neq 0$, then $\exists$ a positive number λ and a positive integer m_3 such that

$$|b_n - b| < \frac{1}{2}|b|, \ \forall \ n \geq m_3,$$

Thus, $\qquad |b| - |b_n| \geq |b_n - b| < \frac{1}{2}|b|.$

$$\Rightarrow \qquad |b_n| \geq \frac{1}{2}|b| \, (say), \ \forall \ n \geq m_3. \qquad (2)$$

Let us apply the Lemma to prove the main theorem.

Now

$$\left|\frac{a_n}{b_n} - \frac{a}{b}\right| = \left|\frac{ba_n - ab_n}{bb_n}\right| = \left|\frac{b(a_n - a) - a(b_n - b)}{bb_n}\right|$$

$$\leq \frac{|b|\,|a_n - a| + |a|\,|b_n - b|}{|b|\,|b_n|}$$

$$\leq \frac{2}{|b_n|}\,|a_n - a| + \frac{2\,|a|}{|b|^2}\,|b_n - b|\,,\ \forall\, n \geq m_3 \qquad [using(2)]$$

Let $\varepsilon > 0$ be given,
Since $\lim a_n = a$ and $\lim b_n = b$, therefore, $\exists$ positive integers m_1, m_2 such that

$$|a_n - a| < \frac{1}{4}\,|b|\,\varepsilon,\ \forall\, n \geq m_1$$

and

$$|b_n - b| < \frac{1}{4}\frac{|b|^2\,\varepsilon}{|a| + 1},\ \forall\, n \geq m_2$$

Thus, for $m = \max(m_1, m_2, m_3)$, we have

$$\left|\frac{a_n}{b_n} - \frac{a}{b}\right| < \frac{1}{2}\varepsilon + \frac{1}{2}\varepsilon = \varepsilon,\ \forall\, n \geq m$$

Hence, $\qquad \lim\left(\frac{a_n}{b_n}\right) = \frac{a}{b} = \frac{\lim a_n}{\lim b_n}.$

$$Q.E.D$$

Selected Sequences

The following are the selected real sequences :

(1) $A = \left(\dfrac{1}{n}\right)_{n\in\mathbb{N}}$

(2) $B = \left(\left(1 + \dfrac{1}{n}\right)^{n}\right)_{n\in\mathbb{N}}$

(3) $C = \left(\dfrac{n}{n+1}\right)_{n\in\mathbb{N}}$

(4) $D = \left(\dfrac{5n+3}{2n+1}\right)_{n\in\mathbb{N}}$

Limits of the Selected Sequences

(1) $A = \left(\dfrac{1}{n}\right)_{n\in\mathbb{N}}$

$a_1 = \frac{1}{1},\ a_2 = \frac{1}{2},\ a_3 = \frac{1}{3},\ \ldots$

$A = (1,\ \frac{1}{2},\ \frac{1}{3},\ \ldots)$

As $n \to \infty$, the denominator n increases without bound, while the numerator remains constant.

Hence, the value of $\dfrac{1}{n}$ tends to Zero.

Therefore,

$$\boxed{\lim_{n\to\infty}\left(\frac{1}{n}\right) = 0}$$

(2) $B = \left(\left(1 + \dfrac{1}{n}\right)^n\right)_{n\in\mathbb{N}}$

$b_1 = 2,\ b_2 = \frac{9}{4},\ b_3 = \frac{64}{27},\ \ldots$

$B = (2,\ \frac{9}{4},\ \frac{64}{27},\ \ldots)$

Let $a = \left(1 + \frac{1}{n}\right)^n$

Taking the natural logarithm of both sides, we obtain,

$ln(a) = ln\left(1 + \frac{1}{n}\right)^n$

$\Leftrightarrow ln(a) = n \cdot ln\left(1 + \frac{1}{n}\right)$ $\qquad [\because ln(a^m) = m \cdot ln(a)]$

Using Taylor series[2], we expand $ln\left(1 + \frac{1}{n}\right)$

for $|x| < 1^a$,

$$ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$$

[a]Since $\frac{1}{n} \to 0$ as $n \to \infty$, we have $\left|\frac{1}{n}\right| < 1$ for all sufficiently large n. Hence the Taylor expansion of $\ln(1 + x)$ is valid for $x = \frac{1}{n}$.

Substituting $x = \frac{1}{n}$, we obtain,

$$ln\left(1 + \frac{1}{n}\right) = \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \frac{1}{4n^4} + \ldots$$

[2]**check Appendix A**

Multiplying both sides by $n > 0$, we obtain,

$$n \cdot ln\left(1 + \frac{1}{n}\right) = 1 - \frac{1}{2n} + \frac{1}{3n^2} - \frac{1}{4n^3} + \dots$$

$$\Rightarrow ln(a) = 1 - \frac{1}{2n} + \frac{1}{3n^2} - \frac{1}{4n^3} + \dots$$

As $n \to \infty$, the right-hand side converges to 1, and hence

$$ln(a) \to 1$$

By continuity of the exponential function[3],

$$a = e^{ln(a)} \to e^1 = e$$

Therefore,

$$\boxed{\lim_{n\to\infty}\left(\left(1 + \frac{1}{n}\right)^n\right) = e\ (\approx 2.718)}$$

[3]**check Appendix B**

(3) $C = \left(\frac{n}{n+1}\right)_{n\in\mathbb{N}}$

$c_1 = \frac{1}{2},\ c_2 = \frac{2}{3},\ c_3 = \frac{3}{4},\ \ldots$

$C = (\frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \ldots)$

Let $c_n = \dfrac{n}{n+1}$

As $n \to \infty$, both the numerator and the denominator tend to infinity, yielding an indeterminate form of type $\frac{\infty}{\infty}$.

To evaluate the limit, we perform the following algebraic manipulations.

Dividing both the numerator and the denominator by $n > 0$, we obtain,

$$c_n = \frac{1}{(1 + \frac{1}{n})}$$

As $n \to \infty$, the denominator $(1 + \frac{1}{n})$ tends to 1, Since $\frac{1}{n}$ tends to zero, while the numerator remains constant.

Therefore,

$$\boxed{\lim_{n\to\infty}\left(\frac{n}{n+1}\right) = 1}$$

Alternatively,

$$\frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \left(\frac{1}{n+1}\right)$$

As $n \to \infty$, $\Rightarrow (n+1) \to \infty$,

$$\Rightarrow \left(\frac{1}{n+1}\right) \to 0$$

Therefore,

$$\boxed{\lim_{n\to\infty} \left(\frac{n}{n+1}\right) = 1}$$

(4) $D = \left(\dfrac{5n+3}{2n+1}\right)_{n\in\mathbb{N}}$

$d_1 = \frac{8}{3},\ d_2 = \frac{13}{5},\ d_3 = \frac{18}{7},\ \ldots$

$D = (\frac{8}{3},\ \frac{13}{5},\ \frac{18}{7},\ \ldots)$

Let $d_n = \dfrac{5n+3}{2n+1}$

As $n \to \infty$, both the numerator and the denominator tend to infinity, yielding an indeterminate form of type $\frac{\infty}{\infty}$.

To evaluate the limit, we perform the following algebraic manipulations.

Dividing both the numerator and the denominator by $n > 0$, we obtain,

$$d_n = \frac{(5 + \frac{3}{n})}{(2 + \frac{1}{n})}$$

as $n \to \infty$, the numerator $(5 + \frac{3}{n})$ tends to 5 and the denominator $(2 + \frac{1}{n})$ tends to 2, Since both $\frac{3}{n}$ and $\frac{1}{n}$ tends to zero.

Therefore,

$$\boxed{\lim_{n\to\infty}\left(\frac{5n+3}{2n+1}\right) = \frac{5}{2}}$$

Verification by $\varepsilon - N$ definition

(1) $\lim_{n\to\infty}\left(\frac{1}{n}\right) = 0$

Using the $\varepsilon - N$ definition of the limit of a sequence, we must show:

> For every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $\left|\frac{1}{n} - 0\right| < \varepsilon$, for all $n \geq \mathbb{N}$.

$\Rightarrow \left|\frac{1}{n} - 0\right| < \varepsilon \quad \Leftrightarrow \quad \left|\frac{1}{n}\right| < \varepsilon \quad \Leftrightarrow \quad \frac{1}{n} < \varepsilon \qquad [\because n \in \mathbb{N}]$

$\Leftrightarrow \quad n > \frac{1}{\varepsilon}$

If we choose $N > \frac{1}{\varepsilon}$, then for this choice of N,

if $n \geq N$, then

$$\left|\frac{1}{n} - 0\right| < \varepsilon$$

Since, for every $\varepsilon > 0$, we can find a number $N = \left\lceil\frac{1}{\varepsilon}\right\rceil + 1$

[Ceiling ensures N is a natural number]

Such that

$$\left|\frac{1}{n} - 0\right| < \varepsilon, \quad \text{for all } n \geq N$$

Hence, by the $\varepsilon - N$ definition of convergence,

$$\boxed{\lim_{n\to\infty}\left(\frac{1}{n}\right) = 0}$$

(2) $\lim_{n\to\infty}\left(\left(1+\frac{1}{n}\right)^n\right) = e\ (\approx 2.718)$

Using the $\varepsilon - N$ definition of the limit of a sequence, we must show:

> For every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $\left|\left(1+\frac{1}{n}\right)^n - e\right| < \varepsilon$, for all $n \geq \mathbb{N}$.

[We will use Sandwich theorem to show the result holds.]

> For $0 < x \leq 1$, the following result[a] holds,
>
> $$x - x^2 \ \leq\ ln(1+x)\ \leq\ x$$
>
> [a] **check Appendix C**

Substituting $x = \frac{1}{n}$, we obtain,

$[\because \text{ For } n \geq 1, \Rightarrow 0 < \frac{1}{n} \leq 1]$

$$\frac{1}{n} - \frac{1}{2n^2}\ \leq\ ln\left(1+\frac{1}{n}\right)\ \leq\ \frac{1}{n}$$

Multiplying by $n > 0$, we obtain,

$$1 - \frac{1}{2n}\ \leq\ n \cdot ln\left(1+\frac{1}{n}\right)\ \leq\ 1$$

Let,

$$a_n = n \cdot ln\left(1+\frac{1}{n}\right) = ln\left(\left(1+\frac{1}{n}\right)^n\right)$$

$[\because ln(x^m) = m{\cdot}ln(x)]$

So, we have,

$$1 - \frac{1}{2n}\ \leq\ a_n\ \leq\ 1$$

$$\Rightarrow \quad e^{1-\frac{1}{2n}} \;\leq\; e^{a_n} \;\leq\; e^1$$

$$[\because \text{ for } 0 < x \leq 1, e^x \text{ is increasing}]$$

So, for all $n \geq 1$,

$$e^{1-\frac{1}{2n}} \;\leq\; \left(1+\frac{1}{n}\right)^n \;\leq\; e$$

Let,

$$b_n = \left(1+\frac{1}{n}\right)^n$$

So,

$$e^{1-\frac{1}{2n}} \;\leq\; b_n \;\leq\; e$$

As $n \to \infty$, $1 - \dfrac{1}{2n} \to 1$

So, by continuity of the exponential function,

$$e^{1-\frac{1}{2n}} \to e$$

Thus, both the lower bound and the upper bound for b_n tends to e.

We have, for all $n \geq 1$:

$$e^{1-\frac{1}{2n}} \;\leq\; b_n \;\leq e$$

So, $b_n \leq e$, and hence

$$|b_n - e| = e - b_n$$

Also,

$$e^{1-\frac{1}{2n}} \leq b_n \quad \Rightarrow \quad e - b_n \leq e - e^{1-\frac{1}{2n}}$$

Thus, for all $n \geq 1$:

$$|b_n - e| \leq e - e^{1-\frac{1}{2n}}$$

Let,

$$c_n = e - e^{1-\frac{1}{2n}}$$

We know,

$$1 - \frac{1}{2n} \to 1 \quad \Rightarrow \quad e^{1-\frac{1}{2n}} \to e \quad \Rightarrow \quad c_n = e - e^{1-\frac{1}{2n}} \to 0$$

So for the sequence c_n, by the **definition of limit**, we have:

> For every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that
> $|c_n - 0| = c_n = e - e^{1-\frac{1}{2n}} < \varepsilon$, for all $n \geq \mathbb{N}$.

Now recall

$$|b_n - e| \leq c_n = e - e^{1-\frac{1}{2n}}$$

So, for that same choice of N, for all $n \geq N$:

$$|b_n - e| \leq e - e^{1-\frac{1}{2n}} < \varepsilon$$

$$\Rightarrow \qquad \left| \left(1 + \frac{1}{n}\right)^n - e \right| < \varepsilon, \quad \text{for all } n \geq N$$

Hence, by $\varepsilon - N$ definition of convergence,

$$\boxed{\lim_{n\to\infty} \left(\left(1 + \frac{1}{n}\right)^n \right) = e}$$

(3) $\lim_{n\to\infty}\left(\frac{n}{n+1}\right) = 1$

Using the $\varepsilon - N$ definition of the limit of a sequence, we must show:

> For every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $\left|\frac{n}{n+1} - 1\right| < \varepsilon$, for all $n \geq \mathbb{N}$.

$$\Rightarrow \quad \left|\frac{n}{n+1} - 1\right| < \varepsilon \quad \Leftrightarrow \quad \left|\frac{-1}{n+1}\right| < \varepsilon \quad \Leftrightarrow \quad \left|\frac{1}{n+1}\right| < \varepsilon$$

$$\Leftrightarrow \quad n + 1 > \frac{1}{\varepsilon}$$

$$\Leftrightarrow \quad n > \frac{1}{\varepsilon} - 1 \qquad [\because n \in \mathbb{N}]$$

If we choose $N > \frac{1}{\varepsilon} - 1$, then for this choice of N,

if $n \geq N$, then

$$\left|\frac{n}{n+1} - 1\right| < \varepsilon$$

Since, for every $\varepsilon > 0$, we can find a natural number $N = \left\lceil \frac{1}{\varepsilon} - 1 \right\rceil + 1$

[Ceiling ensures N is a natural number]

Such that

$$\left|\frac{n}{n+1} - 1\right| < \varepsilon, \quad \text{for all } n \geq N$$

Hence, by the $\varepsilon - N$ definition of convergence,

$$\boxed{\lim_{n\to\infty}\left(\frac{n}{n+1}\right) = 1}$$

(4) $\lim_{n\to\infty}\left(\frac{5n+3}{2n+1}\right)=\frac{5}{2}$

Using the $\varepsilon - N$ definition of the limit of a sequence, we must show:

> For every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $\left|\frac{5n+3}{2n+1}-\frac{5}{2}\right| < \varepsilon$, for all $n \geq \mathbb{N}$.

$$\Rightarrow \quad \left|\frac{5n+3}{2n+1}-\frac{5}{2}\right| < \varepsilon \quad \Leftrightarrow \quad \left|\frac{1}{2(2n+1)}\right| < \varepsilon$$

$$\Leftrightarrow \quad 2n+1 > \frac{1}{2\varepsilon}$$

$$\Leftrightarrow \quad n > \frac{1}{4\varepsilon}-\frac{1}{2} \qquad [\because n \in \mathbb{N}]$$

If we choose $N > \frac{1}{4\varepsilon}-\frac{1}{2}$, then for this choice of N,

if $n \geq N$, then

$$\left|\frac{5n+3}{2n+1}-\frac{5}{2}\right| < \varepsilon$$

Since, for every $\varepsilon > 0$, we can find a natural number
$N = \left\lceil \frac{1}{4\varepsilon}-\frac{1}{2}\right\rceil + 1$

[Ceiling ensures N is a natural number]

Such that

$$\left|\frac{5n+3}{2n+1}-\frac{5}{2}\right| < \varepsilon, \quad \text{for all } n \geq N$$

Hence, by the $\varepsilon - N$ definition of convergence,

$$\boxed{\lim_{n\to\infty}\left(\frac{5n+3}{2n+1}\right)=\frac{5}{2}}$$

Remark:

1. N in $\varepsilon - N$ definition is not unique.

2. Any sufficiently large $N(\varepsilon)$ works.

3. Choosing a slightly larger N avoids edge cases.

Numerical Illustration of Convergence

To numerically illustrate the convergence of the selected real sequences, we compute the values of the sequence for increasing values of n and record the corresponding absolute error $|s_n - L|$.

(1) $A = \left(\dfrac{1}{n}\right)_{n\in\mathbb{N}}$

Table 1 Numerical illustration of the convergence of $A = \left(\dfrac{1}{n}\right)_{n\in\mathbb{N}}$

n	$a_n = \dfrac{1}{n}$	Limit L	$\lvert a_n - L\rvert$
1	1.0000	0	1.0000
10	0.1000	0	0.1000
20	0.0500	0	0.0500
50	0.0200	0	0.0200
100	0.0100	0	0.0100
200	0.0050	0	0.0050
500	0.0020	0	0.0020

$$(2)\ B = \left(\left(1+\frac{1}{n}\right)^n\right)_{n\in\mathbb{N}}$$

Table 2 Numerical illustration of the convergence of $B = \left(\left(1+\frac{1}{n}\right)^n\right)_{n\in\mathbb{N}}$

n	$b_n = \left(1+\frac{1}{n}\right)^n$	Limit L	$\lvert b_n - L\rvert$
1	2.000000	e	7.182818×10^{-1}
10	2.593742	e	1.245394×10^{-1}
20	2.653298	e	6.498412×10^{-2}
50	2.691588	e	2.669380×10^{-2}
100	2.704814	e	1.346800×10^{-2}
200	2.711517	e	6.764706×10^{-3}
500	2.715569	e	2.713308×10^{-3}

$$(3)\ C = \left(\frac{n}{n+1}\right)_{n\in\mathbb{N}}$$

Table 3 Numerical illustration of the convergence of $C = \left(\frac{n}{n+1}\right)_{n\in\mathbb{N}}$

n	$c_n = \dfrac{n}{n+1}$	Limit L	$\lvert c_n - L\rvert$
1	0.500000	1	0.500000
10	0.909091	1	9.090909×10^{-2}
20	0.952381	1	4.761905×10^{-2}
50	0.980392	1	1.960784×10^{-2}
100	0.990099	1	9.900900×10^{-3}
200	0.995025	1	4.975124×10^{-3}
500	0.998004	1	1.996008×10^{-3}

$$(4)\ D = \left(\frac{5n+3}{2n+1}\right)_{n\in\mathbb{N}}$$

Table 4 Numerical illustration of the convergence of $D = \left(\frac{5n+3}{2n+1}\right)_{n\in\mathbb{N}}$

n	$d_n = \frac{5n+3}{2n+1}$	Limit L	$\lvert d_n - L\rvert$
1	2.666667	2.5	1.666667×10^{-1}
10	2.523810	2.5	2.380952×10^{-2}
20	2.512195	2.5	1.219512×10^{-2}
50	2.504950	2.5	4.950495×10^{-3}
100	2.502488	2.5	2.487562×10^{-3}
200	2.501247	2.5	1.246883×10^{-3}
500	2.500500	2.5	4.995005×10^{-4}

These numerical observations support the analytical results we obtained earlier.

Graphical Illustration of Convergence

To visually support the analytical results, we present graphical illustrations of the convergence of the selected sequences. In each case, the discrete terms of the sequence are plotted for increasing values of n, together with the corresponding limit.

(1) $A = \left(\dfrac{1}{n}\right)_{n\in\mathbb{N}}$

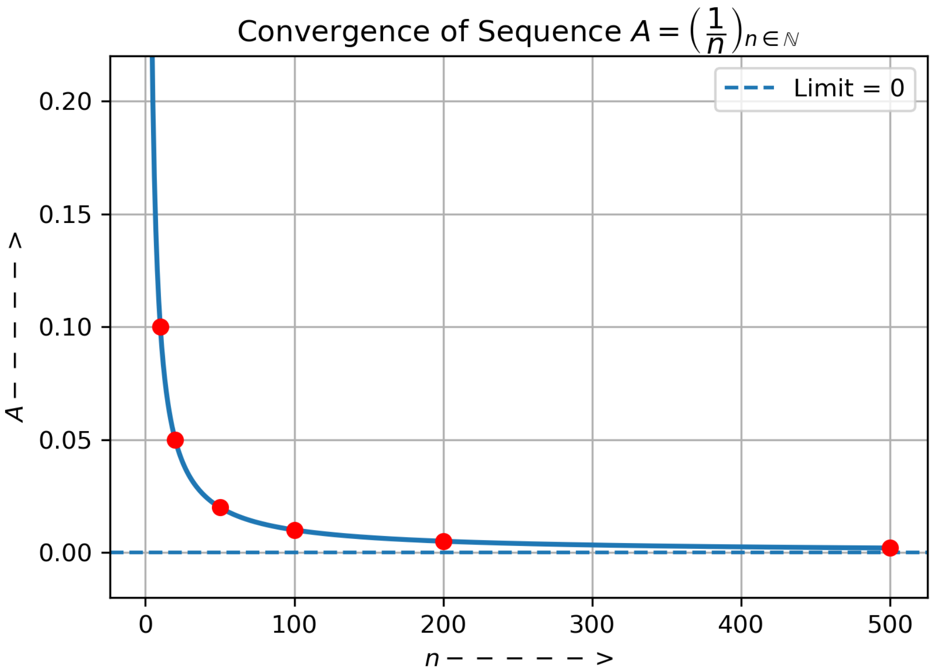

Figure 1

(2) $B = \left(\left(1+\dfrac{1}{n}\right)^n\right)_{n\in\mathbb{N}}$

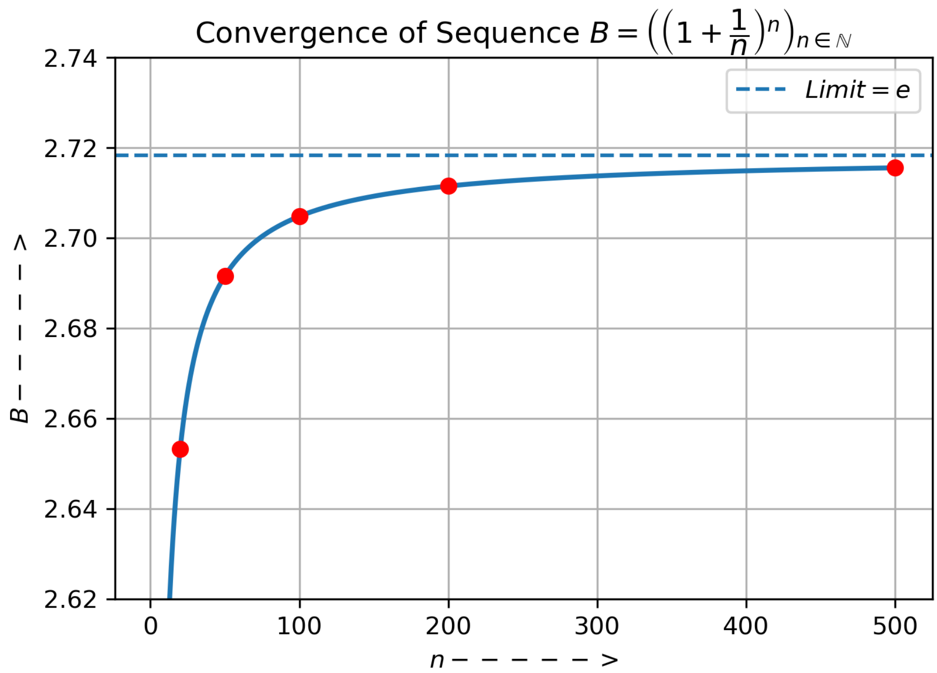

Figure 2

$$(3)\ C = \left(\frac{n}{n+1}\right)_{n\in\mathbb{N}}$$

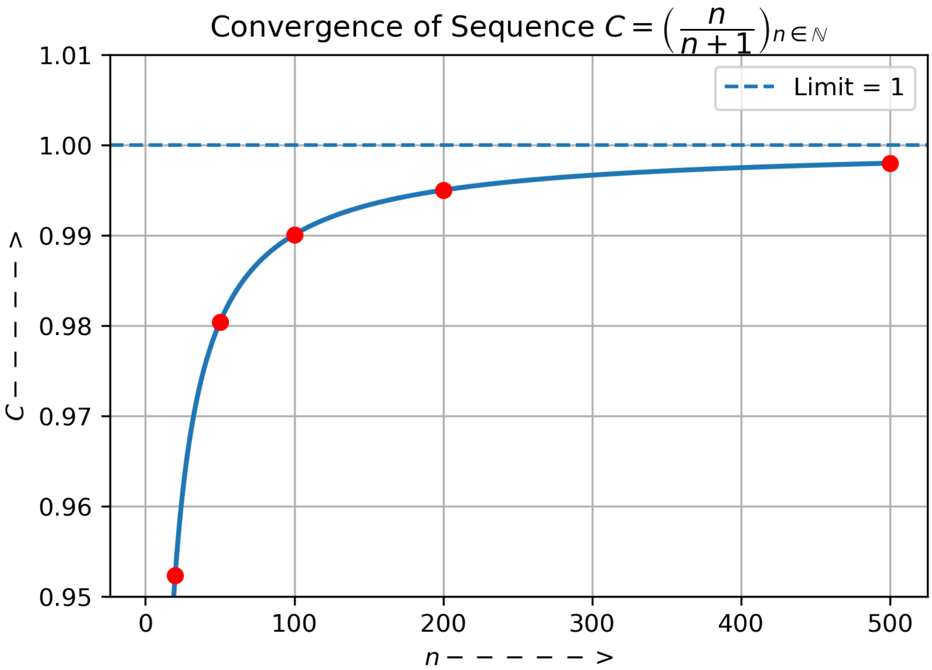

Figure 3

(4) $D = \left(\dfrac{5n+3}{2n+1}\right)_{n\in\mathbb{N}}$

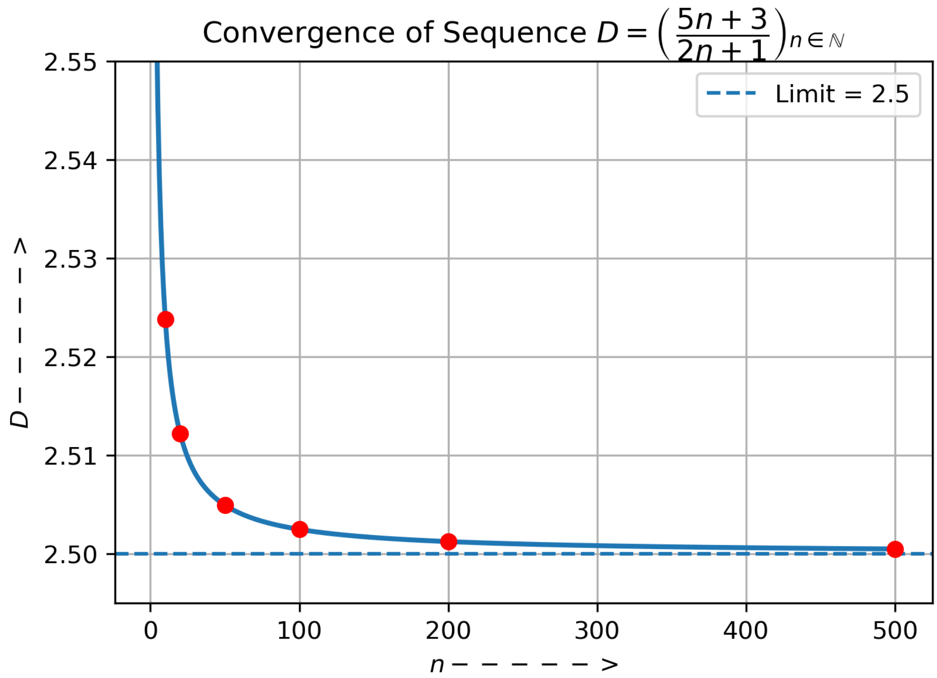

Figure 4

Note:

The smooth curve is included only to guide the eye; the red points represent the actual sequence values.

Discussion

In this project, the convergence of selected real sequences was studied using **analytical proofs, $\varepsilon - N$ verification, numerical approximations, and graphical observations**. All four approaches were found to be consistent and complem–entary, each providing a different perspective on convergence.

The **analytical methods** established the exact limits of the sequences using algebraic manipulation, limit theorems, and known results. These methods gave a rigorous foundation for the conclusion. The $\varepsilon - N$ proofs further confirmed convergence by showing explicitly how large n must be to ensure that the sequence terms remain within a prescribed distance ε from the limit.

The **numerical observations**, obtained through tables of values and error calculations, illustrated how quickly each sequence approaches its limit. It was observed that the seque–nce $a_n = \frac{1}{n}$ converges **slowly** to zero, as the error decreases at a rate proportional to $\frac{1}{n}$. In contrast, the sequence $c_n = \frac{n}{n+1}$ converges more rapidly to 1, since the difference from the limit is $\frac{1}{n+1}$.

A particularly interesting case is the sequence $b_n = \left(1 + \frac{1}{n}\right)^n$, which converges to the constant e. Although its convergence is slower than exponential convergence in a strict sense, it exhibits a distinctive growth-based convergence behaviour compared to the rational sequences. This sequence approaches its limit from below in a monotonic increasing manner, which is clearly visible in the graphical plot.

The **graphical analysis** provided intuitive insight into conv–ergence behaviour. The graphs showed monotonicity, direction of approach (from above or below). In all the cases, the graphical behaviour agreed with the analytical and numerical findings.

Conclusion

In this project, the convergence of four selected real sequences was systematically analyzed. It was shown that the sequences

$$\frac{1}{n},\ \frac{n}{n+1},\ \left(1+\frac{1}{n}\right)^n,\ \text{and}\ \frac{5n+3}{2n+1}$$

all converge as $n \to \infty$, with respective limits $0,\ 1,\ e,\ \frac{5}{2}$.

The convergence of each sequence was established analytically using standard limit techniques and was rigorously verified using the $\varepsilon - N$ **definition of convergence**. Numerical computations and error analysis confirmed the theoretical results and demonstrated how quickly or slowly each sequence approaches its limit. Graphical representations further reinforced these conclusions by providing a visual understanding of convergence behaviour.

One key mathematical insight gained from this study is that **not all convergent sequences approach their limits at the same rate**. Some sequences converge slowly, while others converge relatively faster, even though all satisfy the formal definition of convergence. The project also highlighted the importance of combining analytical rigor with numerical approximations and graphical representations to gain a deeper and more intuitive understanding of convergence.

Overall, this study strengthened the understanding of using multiple complementary approaches in analysis.

References

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[3] Stewart, J. (2016). *Calculus: Early transcendentals* (8th ed.). Cengage Learning.

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Appendix A

Preliminaries on Power Series and Taylor Expansions

A **power series** is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 \ + \ c_1 x \ + \ c_2 x^2 \ + \ c_3 x^3 \ + \ \ldots \qquad (1)$$

where x is a variable and the c_n's are constants called the **coefficients** of the series. For each fixed x, the series (1) is a series of constants that we can test for convergence or divergence. A power series may converge for some values of x and diverge for other values of x. The sum of the series is a function

$$f(x) = c_0 \ + \ c_1 x \ + \ c_2 x^2 \ + \ \ldots \ + \ c_n x^n \ + \ \ldots$$

whose domain is the set of all x for which the series converges. Notice that f resembles a polynomial. The only difference is that f has infinitely many terms.

More generally, a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 \ + \ c_1 (x-a) \ + \ c_2 (x-a)^2 \ + \ \ldots \qquad (2)$$

is called a **power series in $(x - a)$** or a **power series centered at a** or a **power series about a**.

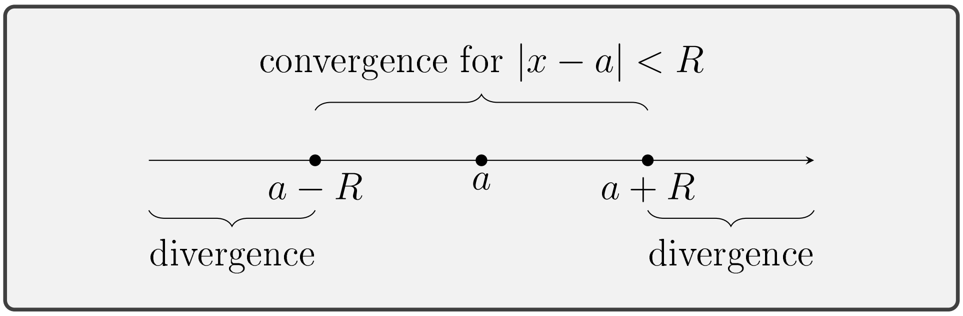
Notice that when writing out the term corresponding to $n = 0$ in Equations 1 and 2, we have adopted the convention $(x-a)^0 = 1$ even when $x = a$. Notice also that when $x = a$, all of the terms are 0 for $n \geq 1$ and so the power series (2) always converges when $x = a$.

> For a given power series $\sum_{n=0}^{\infty} c_n(x-a)^n$, there are only three possibilities:
>
> 1. The series converges only when $x = a$.
> 2. The series converges for all x.
> 3. There is a number $R \in [0, +\infty]$, such that the series converges if $|x - a| < R$ and diverges if $|x - a| > R$.

The number R in case (3) is called the **radius of convergence** of the power series. The **interval of convergence** of a power series is the interval that consists of all values of x.

In case (1) the interval consists of just a single point a. In case (2) the interval is $(-\infty, +\infty)$. In case (3) the inequality $|x - a| < R$ can be rewritten as $a - R < x < a + R$. When x is an *endpoint* of the interval, that is, $x = a \pm R$, anything can happen—the series might converge at one or both endpoints or it might diverge at both endpoints. Thus in case (3) there are four possibilities for the interval of convergence:

$(a - R, a + R)$ $\quad$ $(a-R, a+R]$ $\quad$ $[a-R, a+R)$ $\quad$ $[a - R, a + R]$



Taylor and Maclaurin Series:

Taylor Series, a fundamental concept in mathematics, expresses a **function** as an **infinite sum of terms** which is calculated from the values of its **derivatives at a single point**.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \tag{3}$$
$$= f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ \dots$$

The series in Equation 3 is called the **Taylor series of the function $\boldsymbol{f}$ at $\boldsymbol{a}$** (or **about $\boldsymbol{a}$** or **centered at $\boldsymbol{a}$**). For the special case $a = 0$ the Taylor series becomes

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$$

giving it the special name **Maclaurin series**.

<u>**Remark**</u>: Here, $\quad n! = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot n \quad$ if $n \geq 1$, and $0! = 1$.

Derivation of the Maclaurin Expansion of $\boldsymbol{ln(1+x)}$

Consider the function $f(x) = ln(1+x), \quad x > -1.$

Compute derivatives:

$$\begin{aligned} f(x) &= ln(1+x), \\ f'(x) &= \frac{1}{1+x}, \\ f''(x) &= -\frac{1}{(1+x)^2}, \\ f'''(x) &= \frac{2}{(1+x)^3}, \\ &\vdots \end{aligned}$$

Then evaluate at $x = 0$:

$$\begin{aligned} f(0) &= 0, \\ f'(0) &= 1, \\ f''(0) &= -1, \\ f'''(0) &= 2, \\ &\vdots \end{aligned}$$

$$\begin{aligned} \Rightarrow \;\; f(x) &= f(0) \; + \; \frac{f'(0)}{1!}x \; + \; \frac{f''(0)}{2!}x^2 \; + \; \frac{f'''(0)}{3!}x^3 \; + \; \ldots \\ &= 0 \; + \; \frac{1}{1!}x \; + \; \frac{(-1)}{2!}x^2 \; + \; \frac{2}{3!}x^3 \; + \; \frac{(-6)}{4!}x^4 \; + \; \ldots \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \ldots \end{aligned}$$

Therefore,

$$ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$$

$$OR$$

$$ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \, \frac{x^n}{n}$$

Test for convergence:

We recall,

Ratio Test: (for general series)
If $\sum u_n$ is a series of real numbers, such that

$$\lim_{n\to\infty} \left| \frac{u_{n+1}}{u_n} \right| = L \quad \textbf{(exists)},$$

then the series

(i) converges absolutely, if $L < 1$,

(ii) diverges, if $L > 1$, and

(iii) the test is inconclusive, if $L = 1$.

Let,

$$u_n = (-1)^{n+1} \, \frac{x^n}{n}, \;\Rightarrow\quad u_{n+1} = (-1)^{n+2} \, \frac{x^{n+1}}{n+1}$$

Then,

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{(-1)^{n+2}}{(-1)^{n+1}} \cdot \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right| = \frac{|x|}{1+\frac{1}{n}}$$

$$\Rightarrow \quad \lim_{n\to\infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n\to\infty} \frac{|x|}{1+\frac{1}{n}} = |x|.$$

Therefore, the power series expansion of $ln(1+x)$ is valid for $|x| < 1$.

Appendix B

Exponential and Logarithmic Functions via Power Series

Exponential Function:

We define the exponential function $\exp : \mathbb{R} \to \mathbb{R}$ by

$$\exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}.$$

To verify that this definition is valid for all $x \in \mathbb{R}$, we apply the Ratio Test. Let

$$u_n = \frac{x^n}{n!}.$$

Then

$$\left| \frac{u_{n+1}}{u_n} \right| = \frac{|x|}{n+1} \longrightarrow 0 \quad \text{as } n \to \infty.$$

Hence, the series converges absolutely for all $x \in \mathbb{R}$.

We now define the number e by

$$e := \exp(1).$$

That is,

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

The exponential function $\exp : \mathbb{R} \to \mathbb{R}$ is continuous, strictly increasing, and satisfies

$$\exp(x) > 0 \quad \text{for all } x \in \mathbb{R}.$$

Moreover,

$$\lim_{x \to -\infty} \exp(x) = 0 \quad \text{and} \quad \lim_{x \to \infty} \exp(x) = \infty.$$

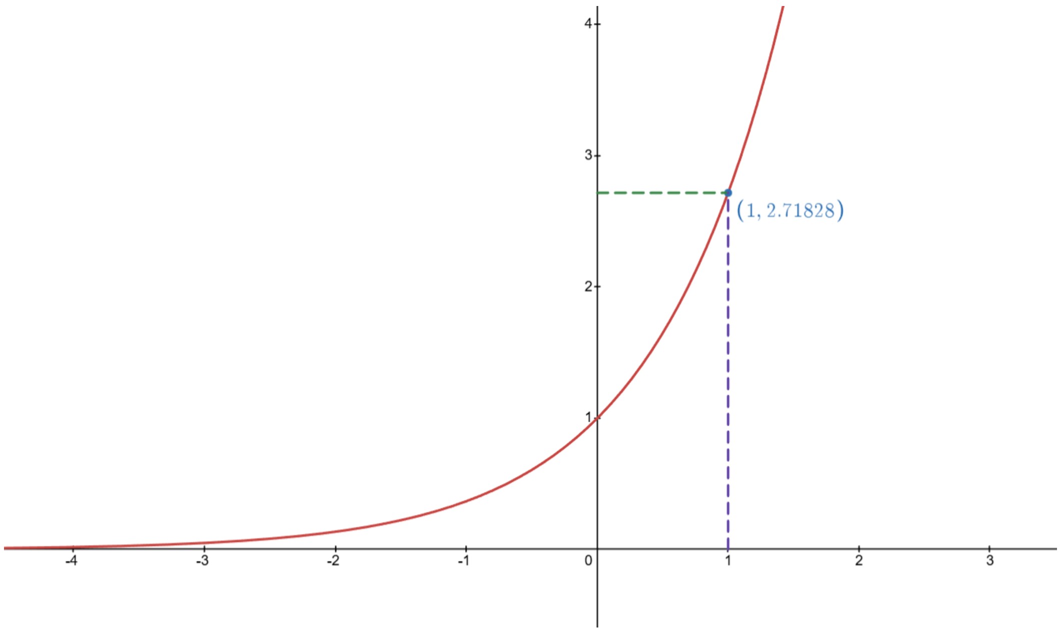


Figure 5 : Graph generated using Desmos Graphing Calculator

Logarithmic Function:

We define the natural logarithm function

$$\ln : (0, \infty) \to \mathbb{R}$$

to be the inverse of the exponential function exp.

That is,

For each $y > 0$, $\ln y$ is defined as the unique real number x such that

$$\exp(x) = y.$$

By definition of the inverse function, we have

$$\ln(\exp(x)) = x \quad \text{for all } x \in \mathbb{R},$$

and

$$\exp(\ln y) = y \quad \text{for all } y > 0.$$

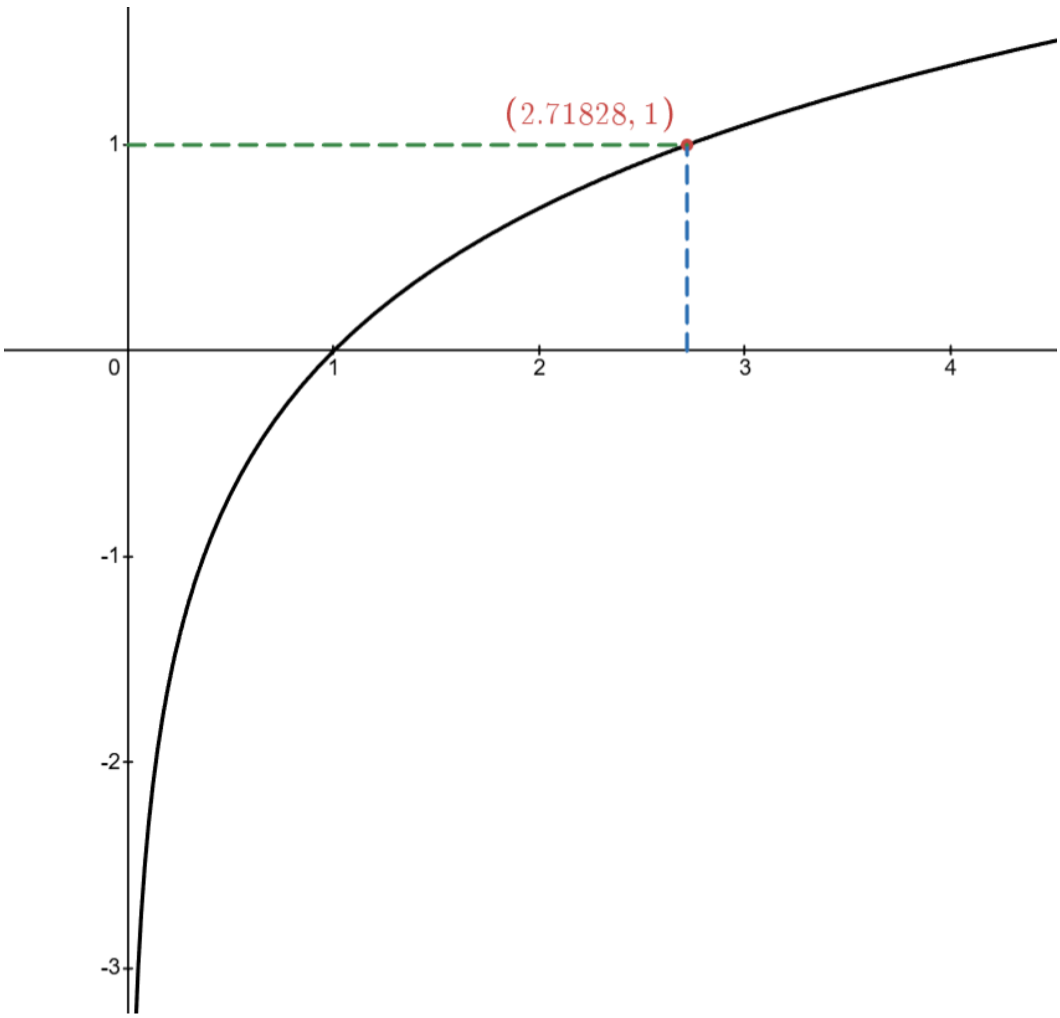


Figure 6 : Graph generated using Desmos Graphing Calculator

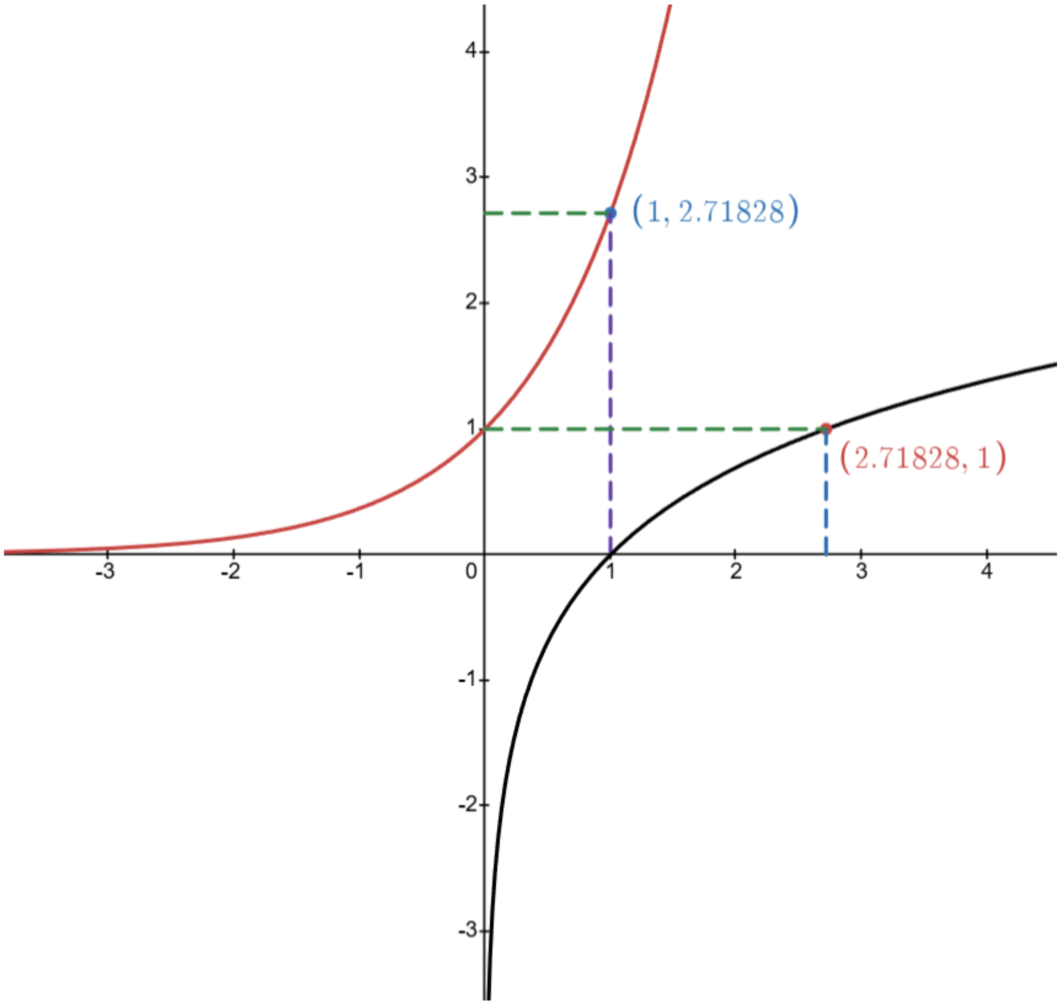


Figure 7 : Graph generated using Desmos Graphing Calculator

- Red graph: $y = exp(x),\ x \in \mathbb{R}$
- Black graph: $y = ln(x),\ x \in (0, \infty)$

Appendix C

Proof of the Inequality

For $0 < x \leq 1$, the following result holds,

$$x - \frac{x^2}{2} \;\leq\; ln(1+x) \;\leq\; x$$

Part - I : $\quad ln(1+x) \;\leq\; x$

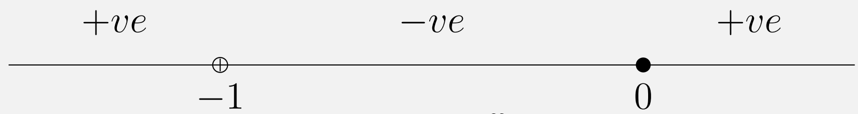
Let us define,

$$g(x) = x - ln(1+x)\,, \quad (x > -1)$$

Differentiating both sides w.r.t x, we obtain,

$$g'(x) = 1 - \frac{1}{1+x} = \frac{1+x-1}{1+x}$$

$$\Rightarrow \quad g'(x) = \frac{x}{x+1}$$



$$g'(x) = \frac{x}{x+1}$$

$\therefore \quad g(x)$ is increasing function in $x \in [0, 1]$

Evaluating at $x = 0$, we obtain,

$\Rightarrow \quad g(0) = 0 - ln(1 + 0) = 0$

Since, $g(x)$ is increasing and $g(0) = 0$, we have,

$$g(x) \geq 0, \quad \forall x \geq 0$$

Hence, for $0 < x \leq 1$,

$$g(x) = x - ln(1 + x) \ \geq \ 0$$

$\boxed{\therefore \quad ln(1 + x) \ \leq \ x}$

Part - II : $\quad x - \dfrac{x^2}{2} \ \leq \ ln(1 + x)$

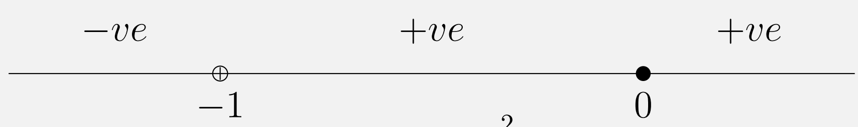
Let us define,

$$h(x) = ln(1 + x) - x + \frac{x^2}{2}, \quad (x > -1)$$

Differentiating both sides w.r.t x, we obtain,

$$h'(x) = \frac{1}{1 + x} - 1 + x \ = \ \frac{1 - (1 + x) + x(x + 1)}{1 + x}$$

$\Rightarrow \quad h'(x) = \dfrac{x^2}{x + 1}$



$$h'(x) = \frac{x^2}{x + 1}$$

$\therefore \quad h(x)$ is increasing function in $x \in [0, 1]$

Evaluating at $x = 0$, we obtain,

$$\Rightarrow \quad h(0) = ln(1+0) - 0 + \frac{0^2}{2} = 0$$

Since, $h(x)$ is increasing and $h(0) = 0$, we have,

$$h(x) \geq 0, \quad \forall x \geq 0$$

Hence, for $0 < x \leq 1$,

$$h(x) = ln(1+x) - x + \frac{x^2}{2} \ \geq \ 0$$

$$\boxed{\therefore \quad x - \frac{x^2}{2} \ \leq \ ln(1+x)}$$

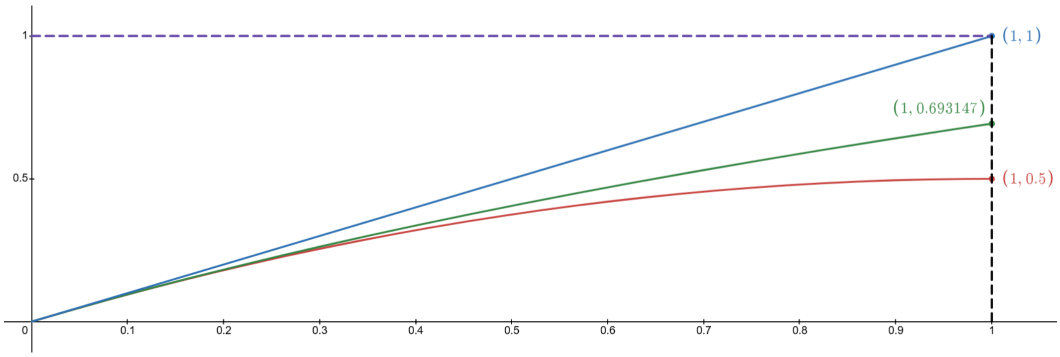


Figure 8 : Graph generated using Desmos Graphing Calculator

- Blue graph : $y = x, \ x \in (0, 1]$
- Green graph : $y = ln(1+x), \ x \in (0, 1]$
- Red graph : $y = x - \dfrac{x^2}{2}, \ x \in (0, 1]$

Appendix D

Additional Graphical Illustrations

(1) $A = \left(\frac{1}{n}\right)_{n\in\mathbb{N}}$

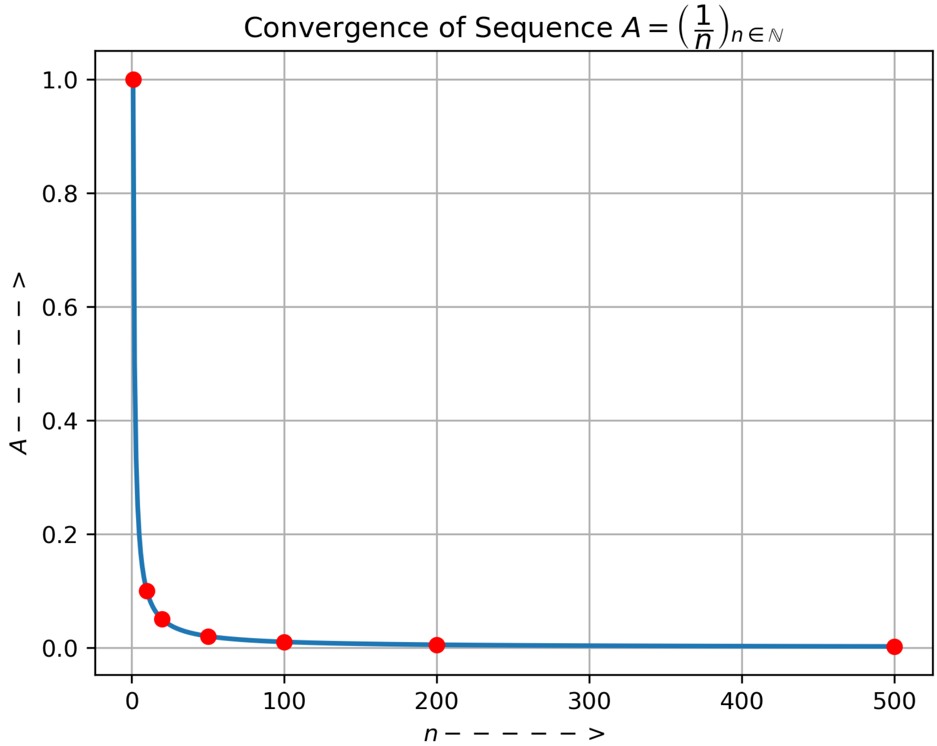


Figure 9

(2) $B = \left(\left(1 + \frac{1}{n} \right)^n \right)_{n \in \mathbb{N}}$

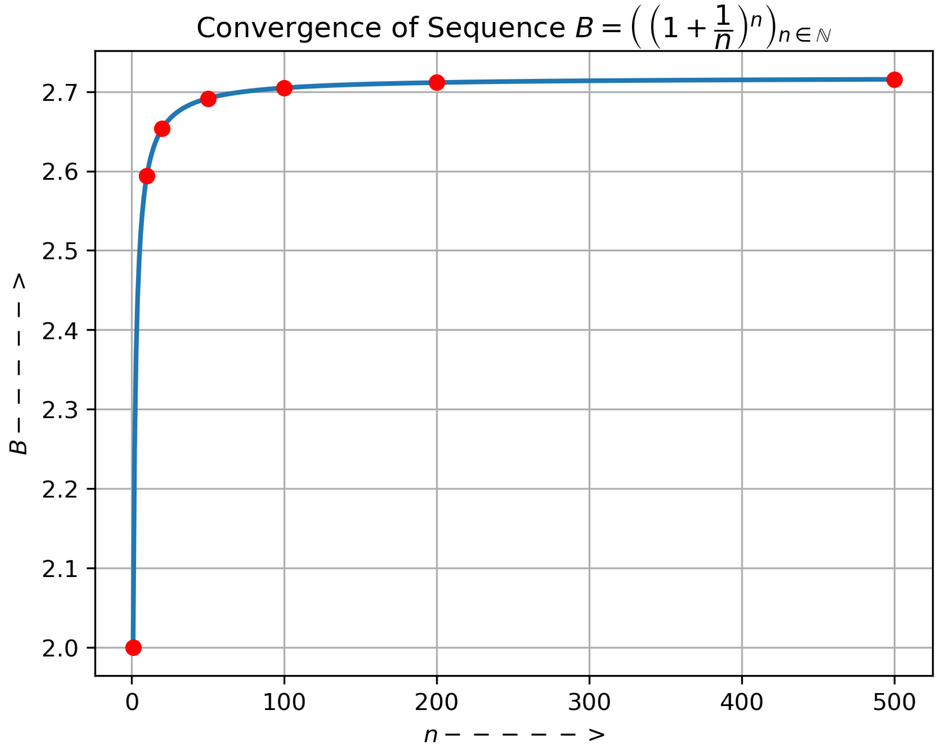


Figure 10

$$(3)\ C = \left(\frac{n}{n+1}\right)_{n\in\mathbb{N}}$$

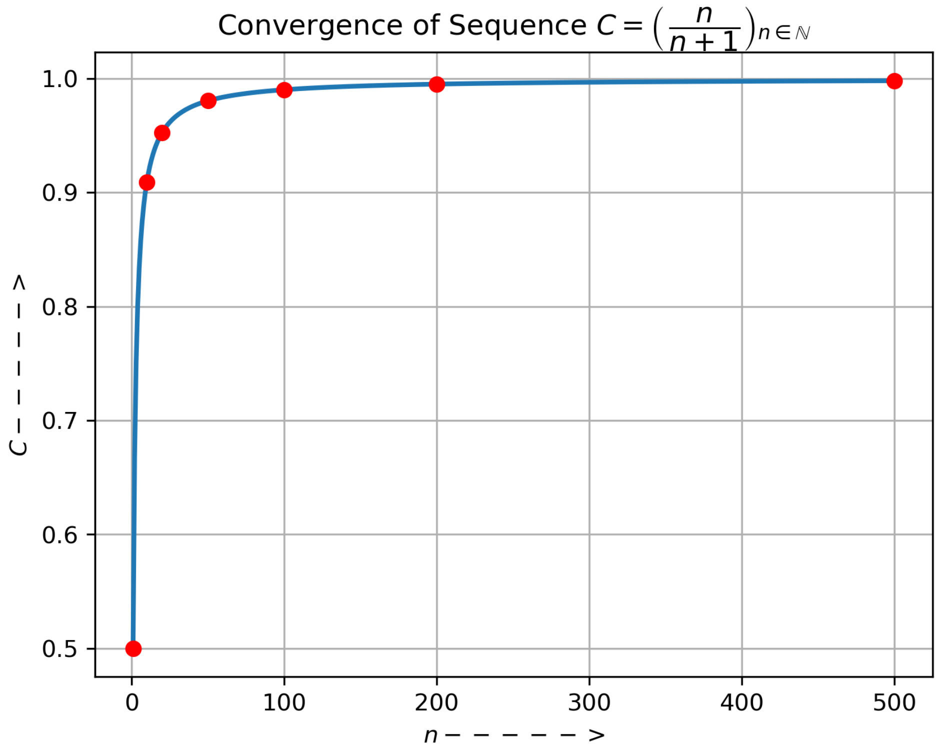


Figure 11

$$(4)\ D = \left(\frac{5n+3}{2n+1}\right)_{n\in\mathbb{N}}$$

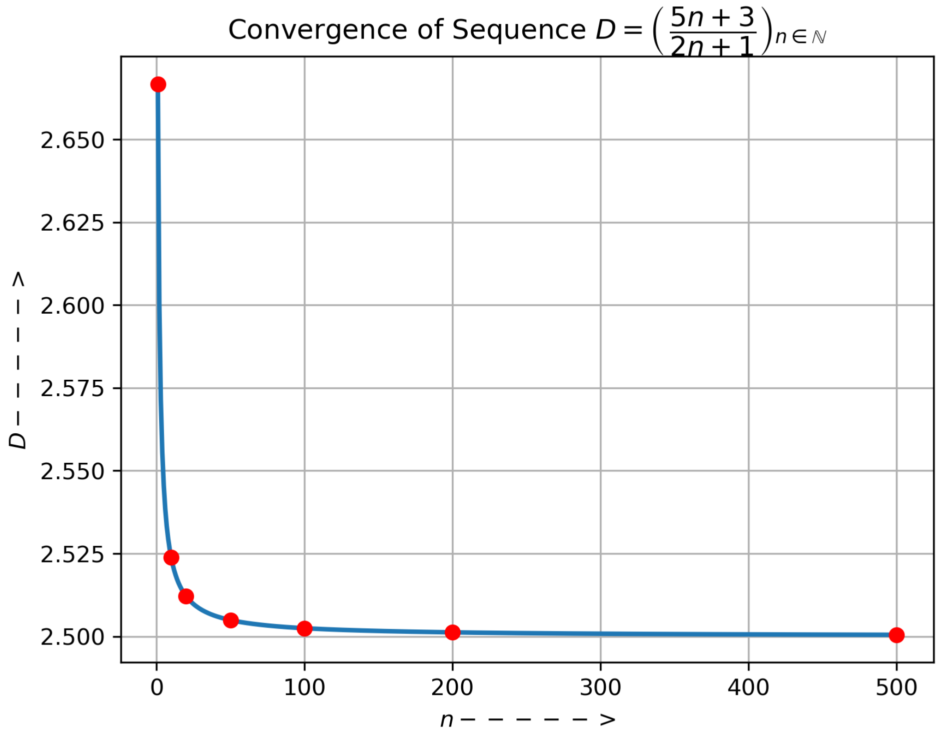


Figure 12

Appendix E

Python Source Code

Numerical computation of sequence values and errors :

```python
import math as m

seq_val = [] # for value of sequence a_n
err_val = [] # for value of error E_n

for n in [1,10,20,50,100,200,500]:
    val = (1/n)
    seq_val.append(val)
    err = m.fabs((1/n) - 0)
    err_val.append(err)

print(f"value of sequence: {seq_val}")
print(f"\nvalue of error: {err_val}")

print("\nValue of Sequence in Scientific Notation: ")
for i in seq_val:
    print(f"{i:.2e}", end = ", ") # converting value of sequence into Scientific Notation with accuracy of 2 decimal places
print("\n\nValue of error in scientific Notation: ")
for i in err_val:
    print(f"{i:.2e}", end = ", ") # converting value of error into Scientific Notation with accuracy of 2 decimal places
```

```python
import math as m

seq_val = [] # for value of sequence a_n
err_val = [] # for value of error E_n

for n in [1,10,20,50,100,200,500]:
    val = (1 + (1/n))**n
    seq_val.append(val)
    err = m.fabs((1 + (1/n))**n - m.e)
    err_val.append(err)

print(f"value of sequence: {seq_val}")
print(f"\nvalue of error: {err_val}")

print("\nValue of Sequence in Scientific Notation with accuracy of 6 decimal places: ")
for i in seq_val:
    print(f"{i:.6e}", end = ", ") # converting value of sequence into Scientific Notation with accuracy of 6 decimal places
print("\n\nValue of error in scientific Notation with accuracy of 6 decimal places: ")
for i in err_val:
    print(f"{i:.6e}", end = ", ") # converting value of error into Scientific Notation with accuracy of 6 decimal places
```

```python
import math as m

seq_val = [] # for value of sequence a_n
err_val = [] # for value of error E_n

for n in [1,10,20,50,100,200,500]:
    val = (n/(n+1))
    seq_val.append(val)
    err = m.fabs((n/(n+1)) - 1)
    err_val.append(err)

print(f"value of sequence: {seq_val}")
print(f"\nvalue of error: {err_val}")

print("\nValue of Sequence in Scientific Notation with accuracy of 6 decimal places: ")
for i in seq_val:
    print(f"{i:.5e}", end = ", ") # converting value of sequence into Scientific Notation with accuracy of 6 decimal places
print("\n\nValue of error in scientific Notation with accuracy of 6 decimal places: ")
for i in err_val:
    print(f"{i:.6e}", end = ", ") # converting value of error into Scientific Notation with accuracy of 6 decimal places
```

```python
import math as m

seq_val = [] # for value of sequence a_n
err_val = [] # for value of error E_n

for n in [1,10,20,50,100,200,500]:
    val = ((5*n + 3)/(2*n + 1))
    seq_val.append(val)
    err = m.fabs(((5*n + 3)/(2*n + 1)) - (5/2))
    err_val.append(err)

print(f"value of sequence: {seq_val}")
print(f"\nvalue of error: {err_val}")

print("\nValue of Sequence in Scientific Notation with accuracy of 6 decimal places: ")
for i in seq_val:
    print(f"{i:.6e}", end = ", ") # converting value of sequence into Scientific Notation with accuracy of 6 decimal places
print("\n\nValue of error in scientific Notation with accuracy of 6 decimal places: ")
for i in err_val:
    print(f"{i:.6e}", end = ", ") # converting value of error into Scientific Notation with accuracy of 6 decimal places
```

Graphical Plots :

```python
import numpy as np
import matplotlib.pyplot as plt

# Creating figure FIRST
plt.figure(figsize=(6, 4))

# Define n values
n = np.arange(1, 501)
a_n = 1 / n

# Plotting the sequence
plt.plot(n, a_n, linewidth=2)
plt.axhline(0, linestyle='--', label="Limit = 0")
plt.ylim(-0.02, 0.22)

# Highlighting selected points
nums = [1, 10, 20, 50, 100, 200, 500]
for Xo in nums:
    Yo = 1 / Xo
    plt.scatter(Xo, Yo, color='red', zorder=5)

# Labels and title
plt.xlabel(r"$n----->$")
plt.ylabel(r"$A----->$")
plt.title(r"Convergence of Sequence $A = \left(\,\dfrac{1}{n}\right)_{n\in\mathbb{N}}$")
plt.grid(True)
plt.legend()
plt.show()
```

```python
import math as m
import numpy as np
import matplotlib.pyplot as plt

# Creating figure FIRST
plt.figure(figsize=(6, 4))

# Define n values
n = np.arange(1, 501)
b_n = (1 + 1 / n)**n

# Plotting the sequence
plt.plot(n, b_n, linewidth=2)
plt.axhline(m.e, linestyle='--', label=r"$ Limit = e $")
plt.ylim(2.62, 2.74)

# Highlighting selected points
nums = [1, 10, 20, 50, 100, 200, 500]
for Xo in nums:
    Yo = (1 + 1 / Xo)**Xo
    plt.scatter(Xo, Yo, color='red', zorder=5)

# Labels and title
plt.xlabel(r"$n----->$")
plt.ylabel(r"$B----->$")
plt.title(r"Convergence of Sequence $B = \left(\left(1 + \dfrac{1}{n}\right)^{n}\right)_{n\in\mathbb{N}}$")
plt.grid(True)
plt.legend()
plt.show()
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Creating figure FIRST
plt.figure(figsize=(6, 4))

# Define n values
n = np.arange(1, 501)
c_n = n / (n + 1)

# Plotting the sequence
plt.plot(n, c_n, linewidth=2)
plt.axhline(1, linestyle='--', label="Limit = 1")
plt.ylim(0.95, 1.01)

# Highlighting selected points
nums = [1, 10, 20, 50, 100, 200, 500]
for Xo in nums:
    Yo = Xo / (Xo + 1)
    plt.scatter(Xo, Yo, color='red', zorder=5)

# Labels and title
plt.xlabel(r"$n----->$")
plt.ylabel(r"$C----->$")
plt.title(r"Convergence of Sequence $C = \left(\,\dfrac{n}{n + 1}\right)_{n\in\mathbb{N}}$")
plt.grid(True)
plt.legend()
plt.show()
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Creating figure FIRST
plt.figure(figsize=(6, 4))

# Define n values
n = np.arange(1, 501)
d_n = (5*n + 3) / (2*n + 1)

# Plotting the sequence
plt.plot(n, d_n, linewidth=2)
plt.axhline(2.5, linestyle='--', label="Limit = 2.5")
plt.ylim(2.495, 2.55)

# Highlighting selected points
nums = [1, 10, 20, 50, 100, 200, 500]
for Xo in nums:
    Yo = (5*Xo + 3) / (2*Xo + 1)
    plt.scatter(Xo, Yo, color='red', zorder=5)

# Labels and title
plt.xlabel(r"$n----->$")
plt.ylabel(r"$D----->$")
plt.title(r"Convergence of Sequence $D = \left(\,\dfrac{5n + 3}{2n + 1}\right)_{n\in\mathbb{N}}$")
plt.grid(True)
plt.legend()
plt.show()
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Define n values
n = np.arange(1, 501)
a_n = 1 / n

# Plotting the sequence
plt.plot(n, a_n, linewidth=2)

# Highlighting selected points
nums = [1, 10, 20, 50, 100, 200, 500]
for Xo in nums:
    Yo = 1 / Xo
    plt.scatter(Xo, Yo, color='red', zorder=5)

# Labels and title
plt.xlabel(r"$n----->$")
plt.ylabel(r"$A----->$")
plt.title(r"Convergence of Sequence $A = \left(\,\dfrac{1}{n}\right)_{n\in\mathbb{N}}$")
plt.grid(True)
plt.show()
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Define n values
n = np.arange(1, 501)
b_n = (1 + 1 / n)**n

# Plotting the sequence
plt.plot(n, b_n, linewidth=2)

# Highlighting selected points
nums = [1, 10, 20, 50, 100, 200, 500]
for Xo in nums:
    Yo = (1 + 1 / Xo)**Xo
    plt.scatter(Xo, Yo, color='red', zorder=5)

# Labels and title
plt.xlabel(r"$n----->$")
plt.ylabel(r"$B----->$")
plt.title(r"Convergence of Sequence $B = \left(\,\left(1 + \dfrac{1}{n}\right)^{n}\right)_{n\in\mathbb{N}}$")
plt.grid(True)
plt.show()
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Define n values
n = np.arange(1, 501)
c_n = n / (n + 1)

# Plotting the sequence
plt.plot(n, c_n, linewidth=2)
# Highlighting selected points
nums = [1, 10, 20, 50, 100, 200, 500]
for Xo in nums:
    Yo = Xo / (Xo + 1)
    plt.scatter(Xo, Yo, color='red', zorder=5)

# Labels and title
plt.xlabel(r"$n----->$")
plt.ylabel(r"$C----->$")
plt.title(r"Convergence of Sequence $C = \left(\,\dfrac{n}{n + 1}\right)_{n\in\mathbb{N}}$")
plt.grid(True)
plt.show()
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Define n values
n = np.arange(1, 501)
d_n = (5*n + 3) / (2*n + 1)

# Plotting the sequence
plt.plot(n, d_n, linewidth=2)

# Highlighting selected points
nums = [1, 10, 20, 50, 100, 200, 500]
for Xo in nums:
    Yo = (5*Xo + 3) / (2*Xo + 1)
    plt.scatter(Xo, Yo, color='red', zorder=5)

# Labels and title
plt.xlabel(r"$n----->$")
plt.ylabel(r"$D----->$")
plt.title(r"Convergence of Sequence $D = \left(\,\dfrac{5n + 3}{2n + 1}\right)_{n\in\mathbb{N}}$")
plt.grid(True)
plt.show()
```

For saving the graphical plots :

```python
plt.savefig("File_Name.png", dpi=300, bbox_inches="tight")
```